

The FREUID Challenge 2026

Technical Report — Brass

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Report date: July 2026

Abstract

We present a domain-adversarial multiple instance learning (DA-MIL) approach to identity document fraud detection. Document images are resized to 672×896 pixels and tiled into a 3×4 grid of 224×224 patches. Each patch is processed by a frozen DINOv2 ViT-B/14 backbone, yielding 3 072 patch tokens per document. Gated attention pooling identifies the most discriminative tokens and produces a bag embedding. A gradient reversal layer suppresses template-specific features during training, pushing the representation towards fraud signals that generalize across unseen document templates. Score calibration applies a sigmoid function directly to the MIL classification head output. The method achieves a FREUID score of **0.068** on the public leaderboard (lower is better). Full reproduction is provided via a Docker image with bundled weights, a GitHub repository (MIT License), and this report.

Competition: The FREUID Challenge 2026 (IJCAI-ECAI 2026) [1]

Kaggle team: Jacopo Bartoli

Code: <https://github.com/JacopoBartoli/freuid-challenge> (commit 23e7b82a612be5d1b492d028fc95cd76d5c48fe0)

1 Introduction

The FREUID Challenge targets fraud detection in identity documents (passports, driving licences). Fraudulent samples span physical alterations and digital manipulations; the threat model encompasses print-and-capture attacks, digital retouching, and generative AI forgeries. A key difficulty is that fraud artefacts are local and visually subtle, while the overall document layout is strongly template-specific. Models that memorize template geometry generalize poorly to the held-out templates used in competition evaluation.

We address both challenges simultaneously: multiple instance learning handles the locality of fraud signals without patch-level annotations, and domain adversarial training prevents the model from exploiting template identity as a shortcut.

2 Method

2.1 Overview

The pipeline has three stages: (1) resolution-preserving tiling, (2) frozen DINOv2 patch encoding, (3) domain-adversarial MIL classification.

2.2 Model Architecture

Tiling. Each input image is resized to 672×896 pixels with bicubic interpolation and partitioned into a 3×4 grid of non-overlapping 224×224 tiles. Resizing to this intermediate resolution preserves significantly more document detail than a direct 224×224 resize.

Feature extraction. Each tile is independently encoded by a **frozen** DINOv2 ViT-B/14 backbone (`vit_base_patch14_dinov2.lvd142m`, pretrained on LVD-142M [1]). With patch stride 14, each tile produces 256 spatial tokens of dimension $d = 768$. The [CLS] token is discarded. The 12 tiles per image are concatenated to form a bag of $N = 3072$ tokens.

Gated Attention MIL. Given bag $\mathbf{H} = \{h_1, \dots, h_N\} \in R^{N \times d}$, attention weights are computed as:

$$a_k = \text{softmax}_k \left(w^\top [\tanh(Vh_k) \odot \sigma(Uh_k)] \right)$$

The bag embedding $z = \sum_k a_k h_k$ is passed to a linear classification head followed by sigmoid to produce a fraud probability in $[0, 1]$. The MIL head has hidden dimension 256 and dropout 0.25, giving approximately 590 K trainable parameters.

Domain adversarial branch. A gradient reversal layer (GRL) [1] placed between the bag embedding z and a 5-class template classifier negates gradients scaled by λ during back-propagation, suppressing template-specific features. The domain branch is inactive at inference time ($\lambda = 0$).

2.3 Training Objective and Procedure

The total loss is $\mathcal{L} = \mathcal{L}_{\text{fraud}} + \mathcal{L}_{\text{domain}}$, where $\mathcal{L}_{\text{fraud}}$ is binary cross-entropy with positive class weight 1.36 and $\mathcal{L}_{\text{domain}}$ is 5-class cross-entropy. The GRL weight follows:

$$\lambda(p) = \lambda_{\max} \cdot \left(\frac{2}{1 + e^{-\gamma p}} - 1 \right), \quad \gamma = 10, \lambda_{\max} = 1.0$$

where $p \in [0, 1]$ is fractional training progress. Only the MIL head receives gradient updates; the DINOv2 backbone is frozen throughout.

2.4 Score Calibration

Raw MIL logits are passed through a sigmoid function, producing a fraud score in $[0, 1]$ with no further post-processing. Higher scores indicate higher confidence of fraud, consistent with the Kaggle submission format.

3 Data

3.1 FREUID Training Set

We use the official FREUID training corpus exclusively. A metadata file mapping images to labels, document templates, and attack types is generated via `scripts/prepare_metadata.py`. We apply **Protocol A**: an 85/15 stratified split by label. The 15% validation split is used for early stopping (best checkpoint by validation FREUID score).

3.2 External Data and Pre-Trained Models

Resource	Source	License	Use
DINOv2 ViT-B/14 (<code>vit_base_patch14_dinov2.lvd142m</code>)	Meta AI / timm LVD-142M pretraining	Apache 2.0	Frozen feature extractor

No external datasets or additional training images were used.

3.3 Validation Strategy

Local validation uses Protocol A (85/15 stratified split). Protocol A is *optimistic* because validation templates overlap with training; local FREUID scores substantially underestimate the true generalization gap. The public leaderboard score (0.068) is our primary performance measure.

4 Inference

At inference time $\lambda = 0$ (domain head inactive). Images are resized to 672×896 , normalized with ImageNet statistics, and tiled as described in Section 2. All 12 tiles per document are passed through the backbone in a single batched forward pass. The attention MIL head aggregates the 3072 tokens and outputs a fraud score. Batch size is 8 documents (96 backbone tile passes per step).

Docker entrypoint.

```
docker run --gpus all --network none \
  -v /path/to/test/images:/data:ro \
  -v /path/to/output:/submissions \
  freuid-inference:latest
```

Output: `/submissions/submission.csv` with columns `id,label` (one row per image, `id` = filename stem). Inference on the public test subset (7821 images) completes in approximately 94 minutes on an NVIDIA GTX 1650 (4 GB VRAM); see Section 6 for the full test set estimate.

5 Results

Table 1: Summary of results. The FREUID score [1] is $1 - 2g_{\text{AuDET}}g_{\text{APCER}}/(g_{\text{AuDET}} + g_{\text{APCER}})$; lower is better.

Method	FREUID ↓	Notes
DA-MIL + tiling, ep 7	0.068	$672 \times 896 \rightarrow 12$ tiles; submitted

The model reaches its best checkpoint at epoch 7, achieving a FREUID score of 0.068 on the public leaderboard.

6 Reproducibility

- **Repository:** <https://github.com/JacopoBartoli/freuid-challenge>, commit 23e7b82a612be5d1b492d028fc95cd76d5c48fe0, MIT License.
- **Checkpoint:** `checkpoints/epoch-7.pt` (1.7MB MIL head only; DINOv2 backbone weights are not stored as they are frozen).
- **Docker:** run `bash docker_build.sh` to build `freuid-inference:latest`. DINOv2 weights are copied from the local HuggingFace cache into the image during build. No network access is required at runtime.
- **Dependencies:** Python 3.10+, PyTorch 2.11, CUDA 12.8, `timm` ≥ 1.0 , `pandas`, `tqdm`, `Pillow`. Full list in `pyproject.toml`.

- **Runtime:** Full test set ($\sim 143\text{K}$ images including private) estimated at ~ 28 hours on GTX 1650 (extrapolated from 94 min/7 821 images); $\sim 4\text{--}5$ hours on a modern mid-range GPU (RTX 3060 or better).
- **Randomness:** No inference-time randomness. Training uses `-seed 42`; DataLoader shuffle is the only non-deterministic element.
- **Network:** Inference container verified with `-network none`; all weights bundled in the image.

Acknowledgments

We thank the FREUID Challenge organizers for preparing the dataset and evaluation infrastructure [1].

References

- [1] Ivan Relić, Vincenzo D’Elia, Stefano Bortoli, Radu Tudoran, Hristina Nedyalkova, Mihaela Bošnjak, Stanislav Pavlić, Tin Mavračić, Darin Dašić, Marin Kačan, Jerko Šegvić, Paolo Čerić, Filip Šoprun, Matej Miočić, Marin Oršić, Lorenzo Vaiani, and Paolo Garza. The FREUID challenge 2026: 1st competition on identity documents fraud detection. <https://freuid2026.microblink.com/>, 2026. IJCAI-ECAI 2026 competition.

A Hyperparameters

Hyperparameter	Value
Backbone	DINOv2 ViT-B/14 (<code>vit_base_patch14_dinov2.lvd142m</code>)
Input resize	672×896 px (bicubic)
Tiling	3×4 grid $\rightarrow$ 12 tiles of 224×224
Tokens per image	$12 \times 256 = 3072$
Token dimension	768
MIL hidden dimension	256
Dropout	0.25
Optimizer	AdamW
Learning rate	1×10^{-4}
LR schedule	CosineAnnealingLR
Fraud loss	BCEWithLogitsLoss, <code>pos_weight = 1.36</code>
Domain loss	CrossEntropyLoss (5-class)
GRL $\lambda_{\max}$	1.0
GRL warmup γ	10
Best checkpoint epoch	7
Protocol	A (85/15 stratified split)
Batch size (training)	16
Batch size (inference)	8 documents
Seed	42

Table 2: Full hyperparameter table.